Simulation of the diocotron instability in pure electron plasmas in planar geometry

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I develop a two-dimensional fluid simulation to study the diocotron instability of a cold, pure electron plasma slab confined between conducting boundaries. The equations of motion for the plasma in this case are isomorphic to the 2D Euler equations for incompressible fluids, and the plasma's flow velocity is dominated by the $\mathbf{E} \times \mathbf{B}$ drift. Introducing sinusoidal perturbation waves on the slab's surfaces, I analyze the growth rate of the waves in the linear regime for different plasma thicknesses and perturbation wavelengths. The simulation results suggest that a thicker slab and a shorter perturbation wavelength lead to a more stable configuration in which the plasma does not evolve into vortices. The peak of the growth rate becomes larger and occurs at smaller thicknesses as the mode number increases, but distinct modes with different perturbation wavelengths share the same stability threshold.

I. INTRODUCTION

The diocotron mode describes an $\mathbf{E} \times \mathbf{B}$ drift wave in low-density, non-neutral plasmas whose space charge's electric field is perpendicular to the external magnetic field [1]. In the case of a cylindrical Penning-Malmberg trap where a pure electron plasma is confined radially by a strong uniform magnetic field along the cylinder axis and axially by charged electrodes at the ends, the $m = 1$ diocotron mode occurs when the plasma column is offset from the axis. This displacement induces image charges on the inner conducting wall; the resulting electric field, in the presence of the axial magnetic field, causes the column to rotate rigidly around the trap. The frequency of this rotation is proportional to the charge density, making the frequency a non-destructive diagnostic for the total confined charge. Theoretical models and experimental efforts to study the diocotron mode, including its stability properties, have been extensive in this geometry, and recent observations of this mode in the more complicated geometry of a dipole magnetic field have motivated further research to derive a dispersion relation that can serve as a diagnostic tool for these confinement systems.

In two levitating dipole traps, A Positron- Electron eXperiment (APEX-LD) and Ring Trap 1 (RT-1), as well as a supporting Lawrence Non-neutral Dipole (LND), signals from wall probes suggest the plasmas propagate toroidally at a frequency consistent with the $m = 1$ $\mathbf{E}\times\mathbf{B}$ rotation frequency [2–4]. Efforts to develop a theoretical model of the diocotron modes in this geometry where the magnetic field is no longer uniform have been attempted in far-field approximations of this problem. Steinbrunner conducted a linear stability analysis on an infinite, straight wire in trying to understand experimental observations in the APEX device [5]. Li simplified the problem further by treating the plasma as a slab in planar geometry, with the external magnetic field homogeneous along the depth of the slab $\mathbf{B} = B\hat{\mathbf{z}}$, through a cold double adiabatic magnetohydrodynamics (MHD) model [6]. Here, I present a two-dimensional fluid simulation of a cold, pure electron plasma slab with Li's configuration and verify his theoretical findings.

In the low density regime, the plasma frequency is negligible compared to the cyclotron frequency $\omega_p^2 \ll \omega_c^2$[7]. As a result, the gyroradius of the cyclotron motion is on a scale that is much smaller than the macroscopic length scale of the guiding centers' drift motion that we are interested in. Futhermore, because the simulation assumes a cold electron plasma ($T = 0$), the plasma's thermal pressure and its associated diamagnetic effects are negligible. This implies the diamagnetic currents do not significantly perturb the external magnetic field in the $z-$direction. Under these conditions, the fluid drift perpendicular to $\mathbf{B}$ is driven entirely by the $\mathbf{E} \times \mathbf{B}$ drift, and a one-fluid MHD model with the guiding-center approximation is sufficient to study the diocotron mode in a planar plasma [8].

When there is shear in the plasma's flow velocity, the plasma breaks up into vortices. This phenomenon is seen in plasmas with a finite thickness such as a hollow electron beam and is referred to as the diocotron instability. This instability resembles the Kelvin-Helmholtz instability seen in fluid dynamics, where electron plasmas are analogous to two-dimensional incompressible fluids [9]. Thus, the equations of motion that govern the low-density electron plasma are isomorphic to the 2D Euler equations for the incompressible fluid. In this paper, building on this analogy, I develop a two-dimensional fluid simulation in Python to study the diocotron instability in an electron plasma slab confined between conducting boundaries. My simulation is executed on a remote server with an Intel Core Ultra 9 Processor 285K CPU model that handles intensive tasks simultaneously using 24 cores.

Simulation results are compared with Li's linear theoretical model. With sinusoidal surface perturbations to the first order applied to the top and bottom surfaces of the slab, we analyze the growth rate of the perturbation waves in the linear regime for different plasma thicknesses and perturbation wavelengths. Qualitatively, both models agree that the diocotron mode is more stable for a thicker slab and a shorter perturbation wavelength. However, a quantitative discrepancy remains between the two models. This could be attributed to the discrete nature of the numerical approximation in my computational model compared to a more continuous treatment in Li's analytical model.

II. SIMULATION MODEL AND METHODS

Figure 1 illustrates the far-field approximation of the dipole established by Li in the three-dimensional Cartesian coordinates. The configuration of the electron plasma looks like a slab of uniform charge density. Under this approximation, the magnetic field $\mathbf{B}$ is straight and uniform along the depth of the slab. This eliminates the complexity of curved magnetic field lines and gradients in the field strength. The plasma length extends in both directions along the $x-$coordinate, and it maintains a finite thickness in the $y-$coordinate.

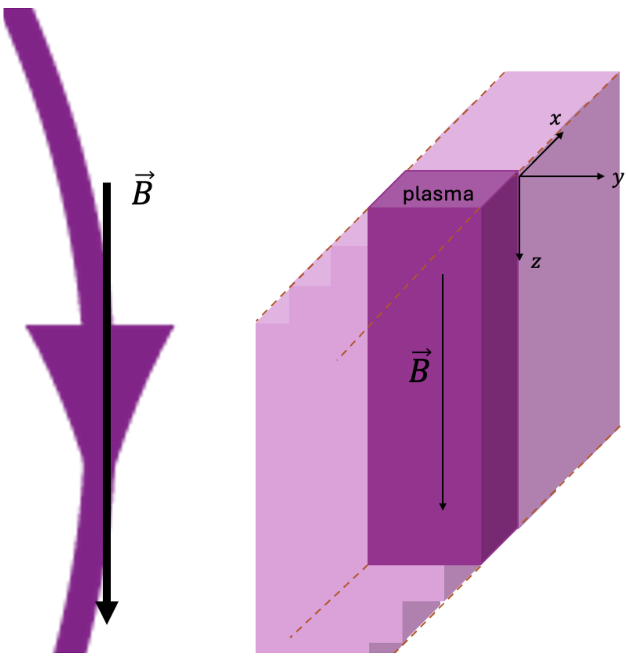

FIG. 1. A far-field approximation of the electron plasma in a dipole magnetic field. Under such approximation, the magnetic field is straight and uniform. In Cartesian coordinates, the plasma is a slab of uniform charge density with a finite thickness in the y-axis, a depth aligned with the direction of the magnetic field in the z-axis, and a length extending periodically along the x-axis.

In the low density regime, the plasma frequency is negligible compared to the cyclotron frequency $\omega_p^2 = \frac{ne^2}{m\epsilon_0} \ll \left(\frac{eB}{m}\right)^2 = \omega_c^2$, where n is the electron density per unit volume, m and e are the mass and charge of an electron respectively, ϵ_0 is the permittivity of free space, and B is the magnetic field strength [7]. This allows us to use the guiding-center approximation and average over the particles' fast gyration on a small scale. Since the guiding center's motion under the diocotron mode is driven entirely by the $\mathbf{E}\times\mathbf{B}$ drift which moves in the direction perpendicular to $\mathbf{B}$, motions along the $z-$coordinate can be neglected. We further reduce the problem from three dimensions to two dimensions by only modeling motions that lie within the $xy-$plane.

A. Simulation setup

The resulting two-dimensional electron plasma slab in the equilibrium state is shown in Fig. 2. I discretize the $xy-$plane using a 200×200 grid with a physical domain of $1\,\text{m} \times 1\,\text{m}$. The magnetic field $\mathbf{B}$ points out of the page in the positive z direction. The plasma with a uniform number density of 10^{12} particles per m^3 is initialized at the center of the grid. The simulation implements periodic boundary conditions along the x-axis and conducting boundary conditions at $y = 0\,\text{m}$ and $y = 1\,\text{m}$.

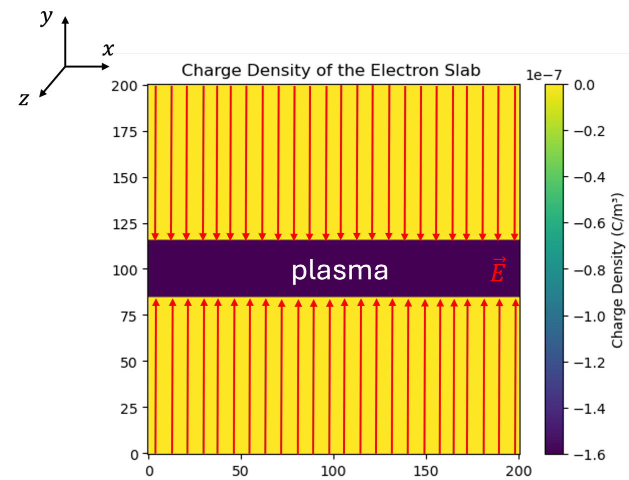

FIG. 2. The equilibrium electron plasma slab is initialized at the center of a 200×200 grid. The physical size of the grid is $1\,\text{m} \times 1\,\text{m}$, and the plasma has a uniform electron density of 10^{12} particles$/m^3$. The top and bottom boundaries are conducting boundaries where the electrostatic potential $\Phi = 0$. The side boundaries are periodic. The electric fields $\mathbf{E}$ above and below the slab point in opposite directions, creating shear in the flow velocity necessary to trigger the diocotron instability.

B. Equations that govern the plasma motion

Given these conditions, the equations of motion for the electron plasma are isomorphic to the 2D Euler equations for incompressible fluids. Thus, we have:

$$\nabla^2\Phi = -\nabla\cdot\mathbf{E} = -\frac{\rho}{\epsilon_0} = \frac{ne}{\epsilon_0}, \tag{1}$$

$$nm_e\left(\frac{\partial}{\partial t} + (\mathbf{u}\cdot\nabla)\right)\mathbf{u} = -\nabla p - ne(\mathbf{E} + \mathbf{u}\times\mathbf{B}), \tag{2}$$

$$\frac{\partial n}{\partial t} + \nabla \cdot (n\mathbf{u}) = 0, \tag{3}$$

$$\mathbf{B} = B\hat{\mathbf{z}}, \tag{4}$$

where Φ, ρ, and p are the electrostatic potential, the charge density, and the kinetic pressure respectively [10]. This treatment of the plasma as a single conducting and charged fluid replaces the individual particles' velocities with the fluid's collective flow velocity $\mathbf{u}$. $\mathbf{E}$ and $\mathbf{B}$ represent the electric field of the plasma's own space charge and the constant external magnetic field pointing in the positive z direction. Equations (1), (2), (3) are Poisson's equation, the fluid equation of motion, and the continuity equation, respectively.

The simulation also assumes a cold electron fluid ($T = 0$) and neglects the electron inertial effect. Consequently, the fluid equation of motion, Eq.(2), reduces to

$$0 = \mathbf{E} + \mathbf{u} \times \mathbf{B} \tag{5}$$

Solving for $\mathbf{u}$, we get

$$\mathbf{u} = \frac{\mathbf{E} \times \mathbf{B}}{B^2} \text{ or } \begin{cases} u_x = \frac{E_y}{B} = -\frac{\partial \Phi}{\partial y}\frac{1}{B} \\ u_y = -\frac{E_x}{B} = \frac{\partial \Phi}{\partial x}\frac{1}{B}. \end{cases} \tag{6}$$

The flow velocity is dominated by the $\mathbf{E} \times \mathbf{B}$ drift. We can also show that the plasma fluid is incompressible in this case by taking the divergence of $\mathbf{u}$:

$$\begin{aligned} \nabla \cdot \mathbf{u} &= \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} \\ &= \frac{\partial}{\partial x}\left(-\frac{\partial \Phi}{\partial y}\frac{1}{B}\right) + \frac{\partial}{\partial y}\left(\frac{\partial \Phi}{\partial x}\frac{1}{B}\right) \\ &= \frac{-\partial^2 \Phi}{\partial x \partial y}\frac{1}{B} + \frac{\partial^2 \Phi}{\partial x \partial y}\frac{1}{B} \\ &= 0. \end{aligned}$$

Eq.(3) then simplifies to

$$\frac{\partial n}{\partial t} + \mathbf{u} \cdot (\nabla \cdot n) = 0 \tag{7}$$

$$\frac{\partial n}{\partial t} + u_x \frac{\partial n_x}{\partial x} + u_y \frac{\partial n_y}{\partial y} = 0. \tag{8}$$

C. Numerical method to solve the partial differential equations

1. *Poisson's equation*

To solve for the electrostatic potential Φ, which arises from the electron density n in Poisson's equation in Eq.(1), the algorithm implements a centered-difference scheme (illustrated in Fig. 3) to replace the second order partial derivative of Φ with algebraic approximations. This method approximates the potential at each grid point $\Phi_{i,j}$ to be the average potential of its four neighboring grid points, known as the solution to the Laplace's equation, plus a term related to the number density at that point,

$$\Phi_{i,j} = \frac{1}{4}(\Phi_{i+1,j} + \Phi_{i-1,j} + \Phi_{i,j+1} + \Phi_{i,j-1}) - (\delta)^2 \frac{e n_{i,j}}{\epsilon_0}, \tag{9}$$

where δ is the uniform grid spacing.

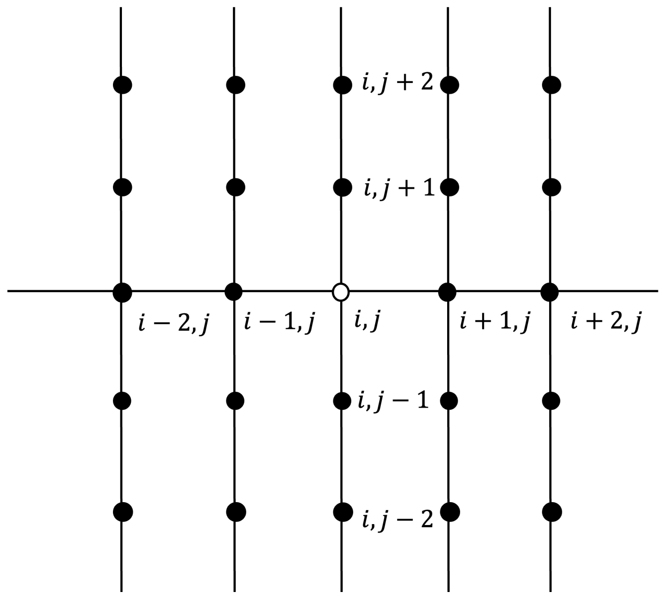

FIG. 3. Discretization of the $xy-$plane using grid points with a uniform spacing of δ. The center-difference method numerically approximates the first order derivative of a function at a grid point by averaging the slope between points on either side of the point of interest $\left(\frac{\partial \Phi_{i,j}}{\partial x} = \frac{\Phi_{i+1,j} - \Phi_{i-1,j}}{2\delta}\right)$.

There are iterative methods such as Jacobi relaxation, Gauss-Seidel, and Successive Over-Relaxation (SOR) methods that solve Eq.(9) at each grid point during the current time step using numerical values of the potentials from the previous time step until the algorithm finds a stable solution that converges. A faster alternative is the matrix-inversion method that places Eq.(9) in the form of $Ax' = b$. Here, A is a sparse $39,600 \times 39,600$ coefficient matrix which represents the linear relationship between the potentials across the grid, excluding 400 grid points on the top and bottom boundaries whose potentials stay fixed at zero; x is a $39,600 \times 1$ vector of the unknown potentials; b is a charge densities vector (Figs. 4 and 5). To solve for x, the algorithm performs a matrix multiplication between the inverse of A and b ($x = A^{-1}b$).

$$A = \underbrace{\begin{bmatrix} T & I & & & I \\ I & T & I & & \\ & I & \ddots & I & \\ & & I & T & I \\ I & & & I & T \end{bmatrix}}_{198\times 200\ cols}\Bigg\}\ 198\times 200\ rows \quad \text{where } I = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{198\ cols}\Bigg\}\ 198\ rows \quad \text{and } T = \underbrace{\begin{bmatrix} -4 & 1 & 0 & 0 & 0 \\ 1 & -4 & 1 & 0 & 0 \\ 0 & 1 & \ddots & 1 & 0 \\ 0 & 0 & 1 & -4 & 1 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix}}_{198\ cols}\Bigg\}\ 198\ rows$$

FIG. 4. Condense form of the coefficient matrix A which represents the linear relationship between the potentials across the grid, excluding the top and bottom boundaries. The sparse matrix A is made up of 198×198 tridiagonal matrices T and identity matrices I. The diagonal entries of T are -4 because the potential at each grid point of interest is the average of its four neighboring grid points, and the off-diagonal entries with a value of 1 correspond to the potentials of the top and bottom neighbors. The identity matrices I directly above and below the main diagonal of the matrix A represent the potentials of the left and right neighbors. The remaining two identity matrices I at the top right and bottom left corners of A represent the periodic boundary conditions along the x-axis.

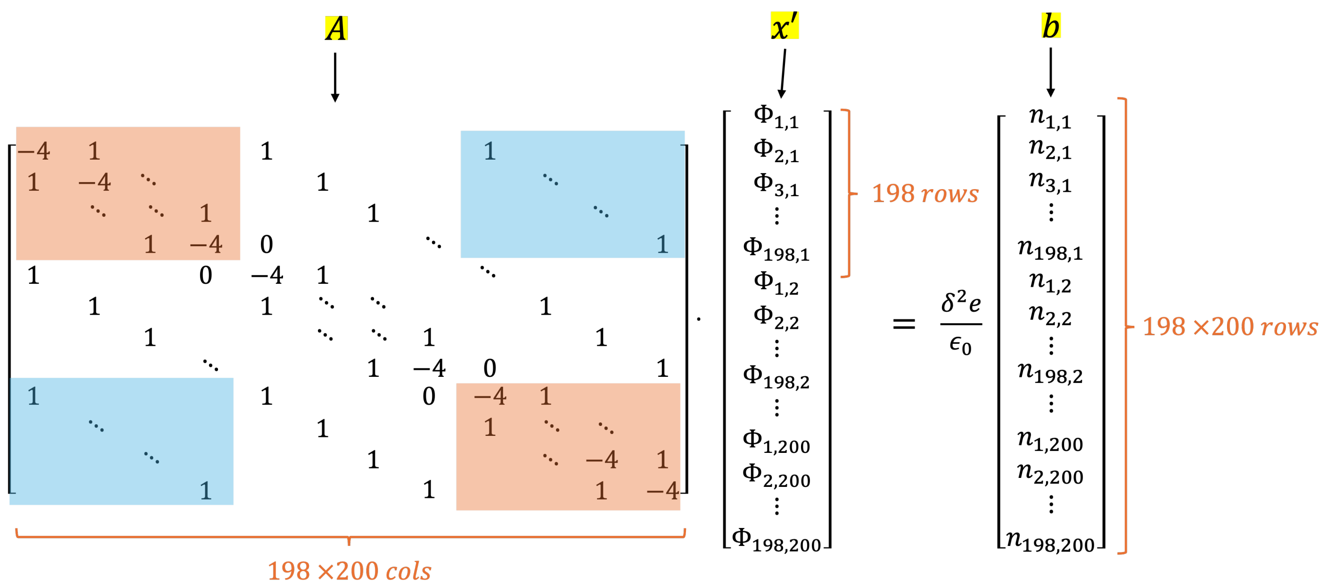

FIG. 5. Equation (9) in the matrix form of $Ax' = b$. A is expanded from Fig. 4 with the orange-shaded boxes representing the tridiagonal matrices T and the blue-shaded boxes representing the identity matrices I. The vector x' shows the flattened 1-dimensional array of the unknown potentials across the grid in an order from top to bottom and left to right. The vector b represents the charge densities at corresponding grid points in x. A and b are known, and the solution for x is given by $x = A^{-1}b$.

2. *Flow velocity*

Once the potential across the entire grid is obtained, the algorithm computes the electric field $\mathbf{E} = -\nabla\Phi$ using a centered-difference approximation. It then "pushes" the fluid in response to the field vector. The numerical treatment of the flow velocity is straightforward from Eq.(6):

$$\begin{cases} u_{x(i,j)} = \frac{E_{y(i,j)}}{B} = -\frac{\Phi_{i,j+1} - \Phi_{i,j-1}}{2\delta}\frac{1}{B} \\ u_{y(i,j)} = -\frac{E_{x(i,j)}}{B} = \frac{\Phi_{i+1,j} - \Phi_{i-1,j}}{2\delta}\frac{1}{B}. \end{cases}$$

3. *Continuity equation*

The resulting Eq.(7) is simplified from the original continuity equation due to the fact that the plasma fluid is incompressible ($\nabla\cdot\mathbf{u} = 0$); however, it is still a partial differential equation that involves both spatial and temporal partial derivatives. The algorithm employs the Forward-Time - Centered-Space (FTCS) method to handle time and space separately. While still discretizing the function spatially using the central difference and assigning the spatial derivatives with the current time step, it uses the current and next time steps to numerically approximate the temporal derivative. Thus, Eq.(7) becomes

$$\frac{n_{i,j}^{t+1} - n_{i,j}^{t}}{\Delta t} + u_{x(i,j)} \frac{n_{i+1,j}^{t} - n_{i-1,j}^{t}}{2\delta} + u_{y(i,j)} \frac{n_{i,j+1}^{t} - n_{i,j-1}^{t}}{2\delta} = 0.$$

D. Initial conditions

The simulation investigates the linear growth of the diocotron instability by introducing sinusoidal perturbation waves on the top and bottom surfaces of the slab (Fig. 6).

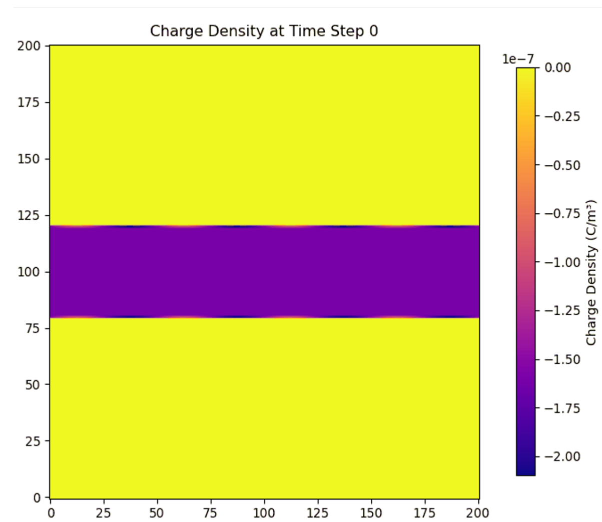

FIG. 6. Sinusoidal perturbation waves applied to the boundaries of the electron plasma slab in its equilibrium state. Subjected to these perturbations, an unstable slab will form vortices as the $\mathbf{E} \times \mathbf{B}$ drifts at the top and bottom surfaces move in opposite directions. The perturbations will only propagate on the surfaces if the slab is stable.

The periodic boundary condition along the $x-$axis restricts the perturbation to integer multiples of the fundamental mode. Consequently, the perturbation's allowed wavelengths are defined as $\lambda = \frac{L}{m}$ and the corresponding wavenumbers are defined as $k = \frac{2\pi}{\lambda} = \frac{2\pi m}{L}$, where $m \in \mathbb{Z}^{+}$ $(m \geq 1)$ and L is the physical length of the grid in $meters$.

The density carries the form:

$$n(x, y) = n_0(x, y) + \tilde{n} \cdot sin(kx), \tag{10}$$

where n_0 represents the equilibrium number density and $\tilde{n}$ represents the amplitude of a small perturbation to the equilibrium.

The length of the grid L is equal to 1 m in both the x and y directions. I initially set up a grid of size 100×100 but increased the size to 200×200 to avoid spatial aliasing. The simulation's time step is on the order of $10^{-10}s$, which is significantly smaller than the inverse of the plasma frequency $(1/\omega_p \sim 10^{-7}s)$. The external magnetic field $\mathbf{B} = 0.1\hat{\mathbf{z}}$ Tesla is initialized at time step 0 and remains constant during the simulation. Again, the number density of the uniform electron plasma slab is 10^{12} particles$/m^3$. Given these parameters, the plasma satisfies the low-density condition with $\omega_p \ll \omega_c$.

III. SIMULATION RESULTS

A. Formation of vortices, a non-linear behavior of the diocotron instability

Figure 7 shows the time evolution of two slabs with identical surface perturbations ($m = 4$ mode) but with different thicknesses for $T = 0,\ 50,\ 100,\ 150,$ and $200\mu s$. As mentioned above, the $\mathbf{E} \times \mathbf{B}$ drift at the top and bottom surfaces moves in opposite directions. In the presence of perturbations, this shear in the flow velocity causes both slabs to eventually break up into vortices. However, the thinner slab in Fig. 7(a) develops into vortices between 50 and 100 μs, while the bottom slab reaches a similar stage after 200 μs.

To observe the effects of different modes m on the stability of the slab, I keep the plasma thickness constant and vary the number of the perturbations' full wavelengths in the initial conditions. In Fig. 8, the plasma half thickness is $d = 0.025m$, or 5 grid cells, for all three slabs. The wavelengths of the surface perturbations are $\lambda = L,\ L/2,$ and $L/4$, which correspond to mode numbers $m = 1,\ 2,$ and 4 respectively. The simulation results show that the slab subjected to perturbations with a longer wavelength is more unstable. Specifically, the $m = 1$ mode forms distinct vortices at $T = 100\mu s$, resulting in earlier plasma loss seen at $T = 200\mu s$, compared to the $m = 4$ mode which does not start to form vortices until after $T = 200\mu s$.

There also seems to be a correlation between the mode number and the number of vortices observed in simulations of thin plasmas or those with a large initial perturbation amplitude. Meanwhile, as shown in Fig. 9, the $m = 1,\ 2,$ and 4 modes for a slab with a half thickness of $d = 0.095m$, or 19 grid cells, all evolve into an identical single-vortex configuration seen at $T = 350,\ 500,$ and $550\ \mu s$ respectively.

To find a stability threshold that is dependent on these two main parameters, the plasma half thickness d and the perturbation mode number m, I vary d in the range between 0.01 and 0.125 m for all three $m = 1,\ 2,$ and 4 modes. The qualitative results shown in Fig. 10 suggest that all three modes share the same stability threshold at $d = 0.1m$. Below this threshold, the plasmas are unstable and break up into vortices. Above this threshold, the perturbation waves only propagate on the surfaces, and

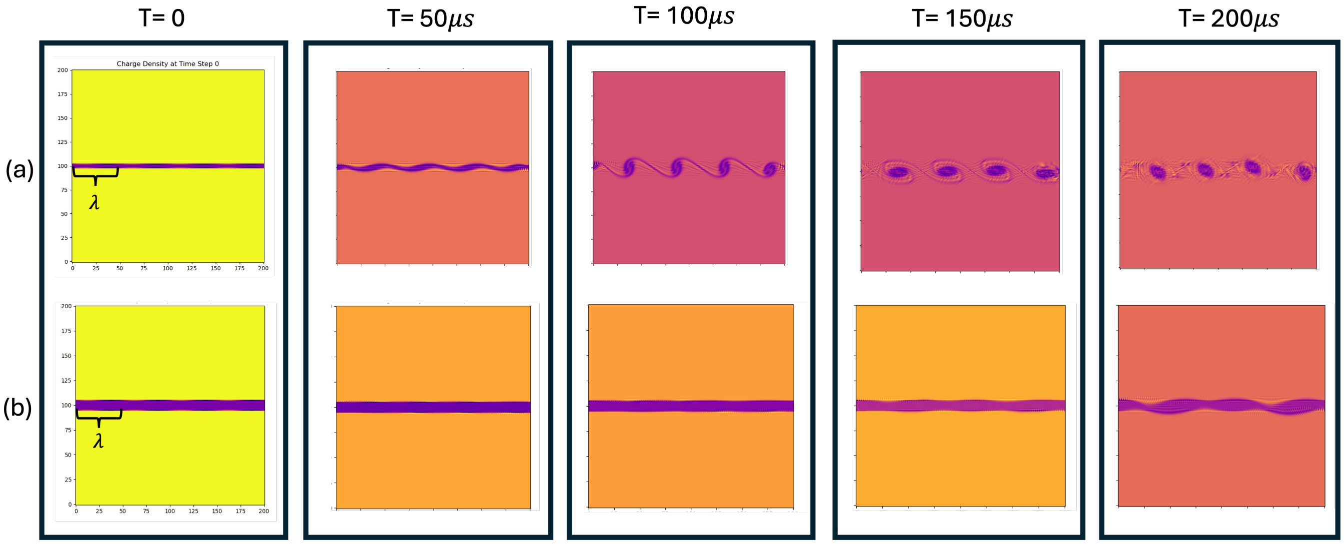

FIG. 7. Time evolution of two slabs with different thicknesses and identical surface perturbations. The slab has a half thickness $d = 0.01m$ or, equivalently, 2 grid cells in (a), and $d = 0.025m$ or, equivalently, 5 grid cells in (b). Both slabs have perturbations with a wavelength of $\lambda = L/4$.

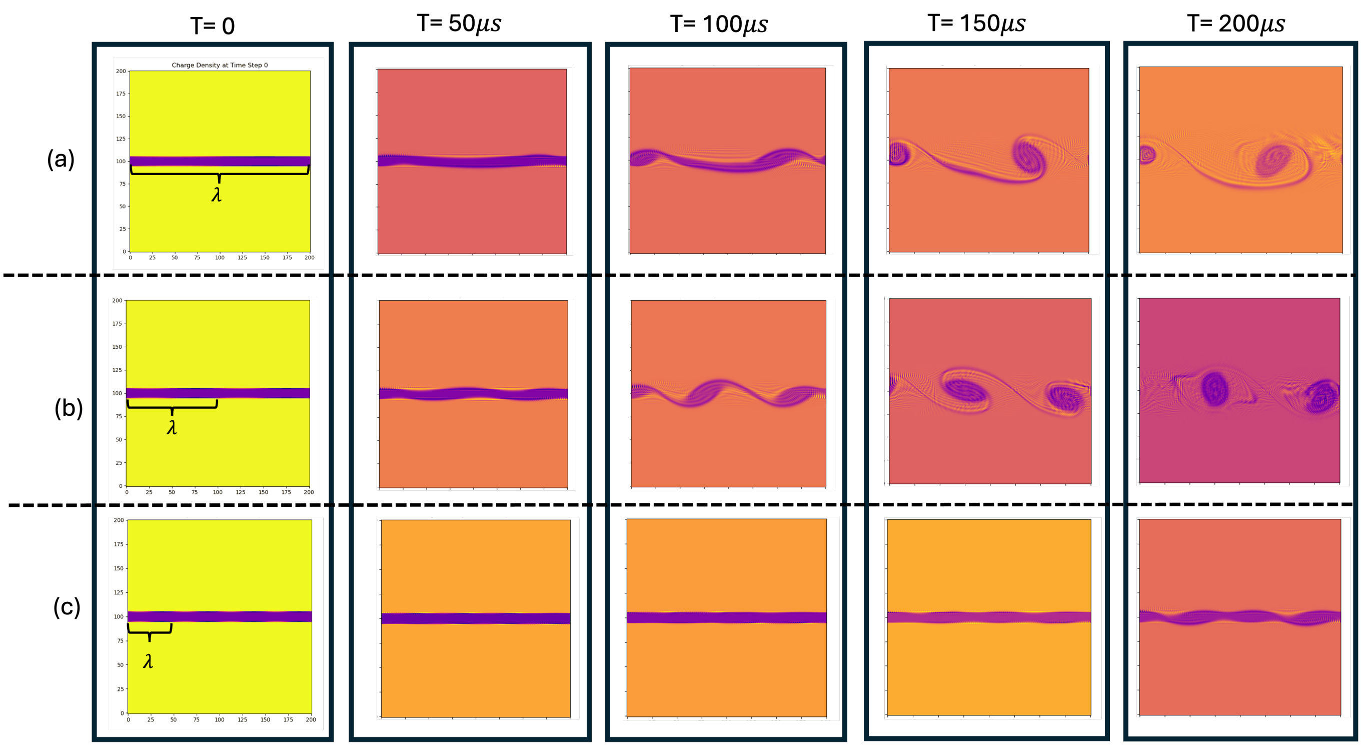

FIG. 8. Time evolution of slabs with the same half thickness $d = 0.025m$ but different perturbation modes $m =$ (a) 1, (b) 2, and (c) 4. The slab in (a) with the longest perturbation wavelength ($m = 1$ mode) is the most unstable; the slab in (c) with the shortest perturbation wavelength ($m = 4$ mode) is the most stable.

the plasmas maintain their initial configurations.

B. Analyzing the linear growth rate of the perturbation waves

Consistent with Li's theoretical model, the qualitative results above indicate that a thicker slab and shorter

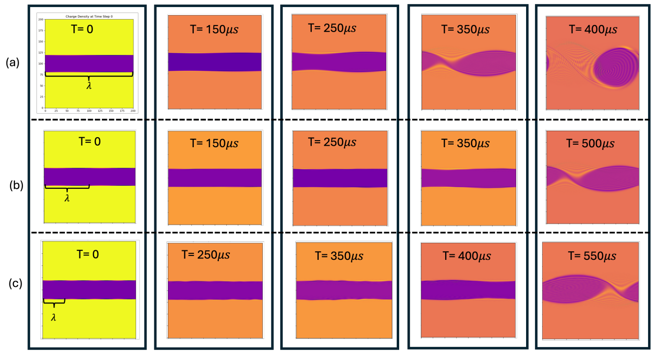

FIG. 9. Time evolution of slabs with a half thickness $d = 0.095m$ or, equivalently, 19 grid cells, under perturbation modes $m =$ (a) 1, (b) 2, and (c) 4.

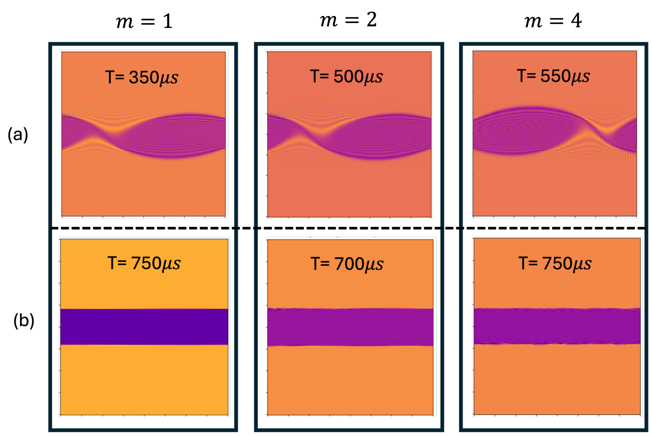

FIG. 10. Stability threshold for different perturbation modes. The plasma half thickness d is varied between 0.01 and 0.125 m for the $m = 1,\ 2,$ and 4 modes. (a) shows the formation of vortices for all three modes at $d = 0.095m$, and (b) shows the absence of vortices for all three modes at $d = 0.1m$. The slabs in (b) are stable, and the perturbation waves only propagate on the surfaces.

perturbation wavelengths lead to a more stable configuration. However, the simulation results deviate from the theoretical predictions regarding the relationship between the stability threshold and the mode number. To examine the simulation more closely, I compute the growth rate of the perturbation waves at the slab's top surface in the linear regime for varying plasma thicknesses and perturbation wavelengths.

During the early time steps, before the slab exhibits the non-linear behavior of the diocotron mode through the formation of vortices, the growth rate of the instability is expected to grow exponentially. This regime is also where analytical models lie within, and the growth rate of the perturbation takes the form:

$$A(t) = A_0 e^{\gamma t}, \tag{11}$$

where $A(t)$ is the amplitude of the perturbation at time t, A_0 is the initial amplitude, which is the same as $n(x, y)$ in Eq.(10), and γ is the growth rate. By taking the natural logarithm of both sides of Eq.(11), we obtain a linear relationship between the logarithm of the perturbation amplitude and time, whose slope is the growth rate γ (Fig. 11).

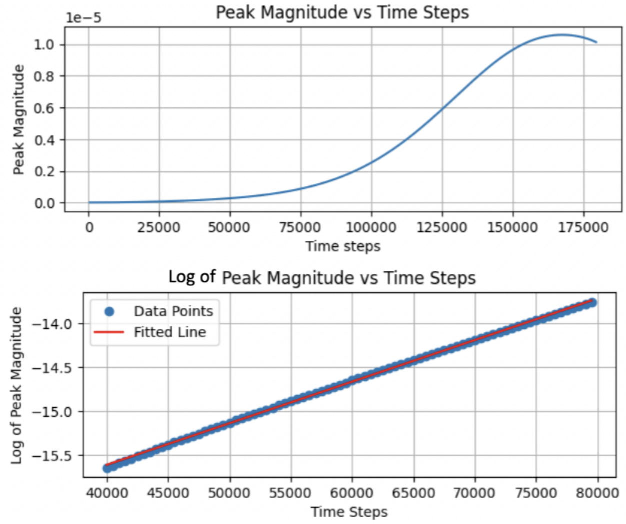

FIG. 11. An example of the exponential growth of the perturbation amplitude in the linear regime (before time step 100,000). By taking the natural logarithm of the amplitude, we obtain a linear relationship between the logarithm of the amplitude and time, whose slope is the growth rate γ.

Figure 12 shows the normalized growth rate $\gamma / \left(\frac{\omega_p^2}{w_c}\right)$ as a function of the plasma's normalized half thickness $d/(0.5m)$ in the unstable region for the $m = 1$ mode. The point at which the normalized growth rate approaches zero ($d_{normalized} = 0.2$) indicates the stability threshold, which is consistent with the qualitative result shown in Fig. 10. The instability growth rate peaks at $d_{normalized} = 0.12$. Beyond the stability threshold, it is harder to analyze the growth rate of the perturbations, but the qualitative results confirm that slabs with a normalized half thickness greater than 0.2 are stable and their perturbations only propagate on the surfaces.

This result qualitatively agrees with Li's result, which also claims that the diocotron mode is more stable for a thicker slab. However, quantitatively, Li found the stability threshold to be at $d_{normalized} = 0.32$ and the peak of the instable growth rate to be at $d_{normalized} = 0.15$ for his lowest mode, $k = 1$ where k is the wave number of his perturbation. This discrepancy could be due to the discrete nature of the numerical approximation and the restriction on the perturbation's wavelength imposed by the periodic boundary conditions in my computational model compared to a more continuous analysis in Li's theoretical model. In the context of the simulation, the mode m refers to the number of full wavelengths that

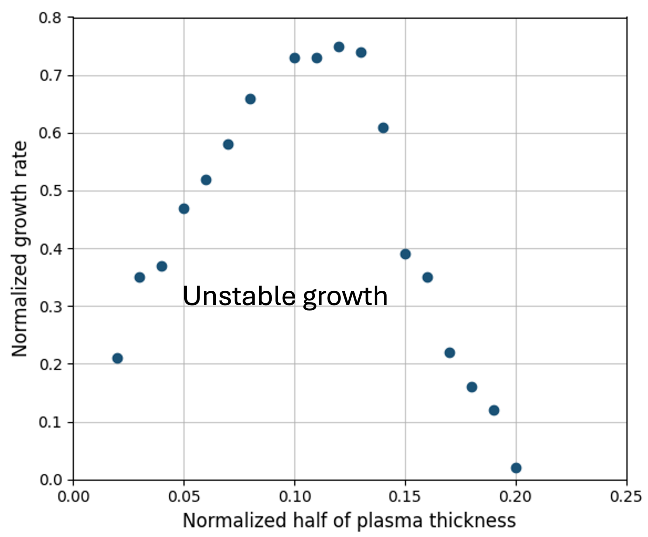

FIG. 12. Normalized growth rate of the perturbation waves $\gamma/\left(\frac{\omega_p^2}{w_c}\right)$ as a function of the plasma normalized half thickness $d/(0.5m)$ in the unstable region for the $m = 1$ mode. The stability threshold which separates the unstable and stable regions is at $d_{normalized} = 0.2$.

fit within the physical length of the grid. Li instead investigated the diocotron mode for an integer number of wavenumber $k = \frac{2\pi}{\lambda}$.

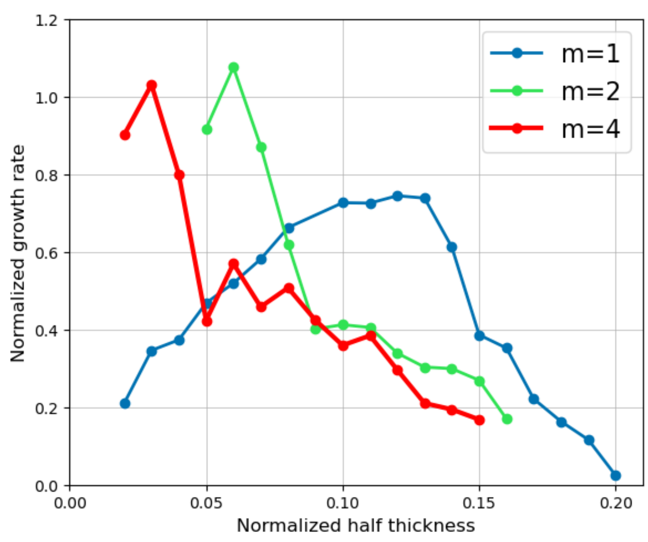

FIG. 13. Normalized growth rate of the perturbation waves as a function of the plasma's normalized half thickness in the unstable region for the $m = 1,\ 2,$ and 4 modes.

In Fig. 13, I investigate the relationship between the perturbation wavelength and its growth rate across different thicknesses in the unstable region. The $m = 1$ mode has the lowest and rightmost instability peak at $d_{normalized} = 0.12$. As the mode number increases, the peak shifts higher and to the left, with the $m = 2$ mode's instability peak at $d_{normalized} = 0.06$ and the $m = 4$ mode's instability peak at $d_{normalized} = 0.03$. This agrees with Li's result that the instability growth rate dependence on the plasma half thickness is specific to the wavenumber of the perturbation. Nevertheless, he observed that as the wavenumber increases and the wavelength decreases, the range of instability shrinks. In his model, different modes have different stability thresholds, with the threshold for the lowest mode being the largest.

IV. CONCLUSION

The two-dimensional fluid simulation presented here provides insights into the behaviors of the diocotron instability in a planar electron plasma slab confined between conducting boundaries. The simulation results qualitatively agree with Li's linear theoretical model regarding the stability dependence of the plasma thickness and perturbation wavelength. In both models, a thicker slab or one exposed to a shorter surface perturbation is more stable. The stability threshold for the plasma half thickness divides the stable and unstable regions. In the unstable region below the stability threshold, the amplitude of the perturbation grows exponentially, causing the slab to break up into vortices. The growth rate peaks at a certain thickness and decreases to zero at the stability threshold, after which the perturbations only propagate on the surfaces.

To investigate the stability properties dependent on the perturbation wavelength, I examine three different modes: $m = 1,\ 2,$ and 4. The simulation suggests that the slab with the longest perturbation wavelength ($m = 1$ mode) develops into vortices and exhibits plasma loss much earlier than the slabs with shorter perturbation wavelengths (higher-order modes). There is a leftward shift in the instability peak as the mode number increases. At smaller plasma half thicknesses, the growth rate of the perturbation waves is larger for higher-order modes, but as the plasma half thickness increases, the lower-order modes dominate the instability growth rate. This could explain the correlation between the number of vortices and the mode number observed in thin plasmas and the convergence into a similar configuration with a low number of vortices in thicker plasmas. Because the $m = 1$ mode has the highest growth rate at larger plasma half thicknesses, the slabs with a half thickness $d = 0.095m$ under all three modes in Fig. 9 evolve into a single vortex, which is an expected behavior of the $m = 1$ mode.

Despite this leftward shift in the instability peak in higher-order modes, the simulation results imply that all three modes share the same stability threshold at $d = 0.1m$. This disagrees with Li's theoretical model, which predicts that the range of instability becomes smaller, equivalent to a smaller stability threshold, as the mode number increases.

The poor agreement between the simulation and the theoretical model could arise from 1) the definition of the mode number in the simulation refers to the discrete number of full wavelengths that fit within the physical length of the grid, constrained by the periodic boundary conditions, as opposed to the continuous wavenumber in Li's model, 2) the crude centered-difference approximation implemented to solve the partial differential equations, and 3) the subjectivity involved in selecting the time interval within the linear regime in the simulation when analyzing the perturbation growth rates. Thus, future work could explore the numerical stability of the centered-difference approximation in this system of equations, as well as more sophisticated numerical methods, including some I encounter during my research such as the Implicit Euler or the Crank-Nicolson methods.

Again, the goal of this project is to understand and derive a dispersion relation for the diocotron instability in experimental non-neutral plasma trapping devices that utilize a dipole magnetic field. This simulation model provides a starting point to study this problem by approaching it from a simplified geometry with idealized conditions. A direct extension of this model is to transition into a geometry more representative of the experimental setup, where the magnetic field is inhomogeneous and curved, such as a straight wire with inner and outer conducting boundaries. While the underlying equation framework remains the same, the simulation would need to be implemented in cylindrical coordinates rather than Cartesian coordinates. One could also investigate the stability properties of the diocotron mode with respect to other parameters such as the distance between the plasma surface and the conducting wall, which is expected to have a stabilizing effect on the instability growth rate.

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